Group Topic: Big Data Analytics for Student Dropout Prediction in Higher Education

Lab No.Deliverable 3

Git Repo/ Colab Link: https://github.com/prinxe0/Big-Data-Analytics-for-Student-Dropout-Prediction-in-Higher-Education

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Objectives

The objective of this activity is to implement machine learning algorithms using big data frameworks to predict student dropout risk. Specifically, the task aims to transition from a rule-based scoring system to a fully trained classification model utilizing the preprocessed and normalized Student Mental Health and Burnout dataset.

Tasks Completed

1. Implementation of Algorithm in code
2. Integrated PySpark MLlib to build and execute the machine learning pipeline.
3. Configured a feature transformer to combine the previously cleaned and scaled numeric features (e.g., stress_level, burnout_score, mental_health_index) into a single unified feature vector.

Tasks to be Completed

1. Model Evaluation and Tuning: Perform tuning and cross-validation to optimize the accuracy, precision, and recall of the trained algorithm.

2. External Tool Integration: Complete the wrapping of the PySpark pipeline into an orchestration tool (such as Apache Airflow) and finalize MLflow metric logging as planned in the previous deliverable.

Results and Analysis

The implementation of the predictive algorithm was successfully executed using PySpark MLlib. Building upon the distributed data preprocessing phase from Deliverable 2, the cleaned dataset was seamlessly ingested into the machine learning pipeline. The dynamically normalized features, specifically the identified primary risk indicators (burnout and stress) and protective indicators (mental health index) were successfully fed into the algorithm. Initial code execution demonstrates that the distributed model successfully compiles and maps the complex relationships between the mental health indicators and the target dropout risk, effectively advancing the project from exploratory data analysis to active predictive modeling.

Challenges and Solutions

Challenge: Formatting the preprocessed dataset to be mathematically compatible with Spark MLlib’s machine learning algorithms, which inherently require all distinct input features to be consolidated into a single vector column for training.

Solution: Implemented PySpark's VectorAssembler functionality to automatically aggregate the disparate, normalized numeric features (such as sleep hours and depression scores) into a unified feature vector prior to fitting the model, ensuring seamless distributed training across the local cores.

Conclusion

The core objective for this deliverable was successfully met. We implemented and executed a machine learning algorithm in code using big data techniques, leveraging the massive dataset prepared in the previous phase. By successfully training a distributed classification model on the mental health indicators, the data engineering pipeline is now capable of automated dropout prediction, setting the stage for final model evaluation and capstone deployment.